Children's Health Care

ISSN: 0273-9615 (Print) 1532-6888 (Online) Journal homepage: www.tandfonline.com/journals/hchc20

A Case Series of Cognitive-Behavioral Therapy Augmentation of Antidepressant Medication for Anxiety in Children With Autism Spectrum Disorders

Eric A. Storch, Joshua M. Nadeau, Brittany Rudy, Amanda B. Collier, Elysse B. Arnold, Adam B. Lewin, P. Jane Mutch & Tanya K. Murphy

To cite this article: Eric A. Storch, Joshua M. Nadeau, Brittany Rudy, Amanda B. Collier, Elysse B. Arnold, Adam B. Lewin, P. Jane Mutch & Tanya K. Murphy (2015) A Case Series of Cognitive-Behavioral Therapy Augmentation of Antidepressant Medication for Anxiety in Children With Autism Spectrum Disorders, Children's Health Care, 44:2, 183-198, DOI: 10.1080/02739615.2014.906310

To link to this article: https://doi.org/10.1080/02739615.2014.906310

Published online: 16 Apr 2015.

Children's Health Care, 44:183–198, 2015

ISSN: 0273-9615 print/1532-6888 online
DOI: 10.1080/02739615.2014.906310

A Case Series of Cognitive-Behavioral Therapy Augmentation of Antidepressant Medication for Anxiety in Children With Autism Spectrum Disorders

Eric A. Storch

Department of Pediatrics, Department of Psychiatry and Behavioral Neurosciences, and Department of Psychology, University of South Florida, Tampa, FL and Rogers Behavioral Health - Tampa Bay, Tampa, FL
All Children's Hospital/John's Hopkins Medicine, St. Petersburg, FL

Joshua M. Nadeau, Brittany Rudy, Amanda B. Collier, and Elysse B. Arnold

Department of Pediatrics, University of South Florida, Tampa, FL

Adam B. Lewin

Department of Pediatrics, Department of Psychiatry and Behavioral Neurosciences, and Department of Psychology, University of South Florida, Tampa, FL

P. Jane Mutch

Department of Pediatrics, University of South Florida, Tampa, FL

Tanya K. Murphy

Department of Pediatrics and Department of Psychiatry and Behavioral Neurosciences, University of South Florida, Tampa, FL

Correspondence should be addressed to Eric A. Storch, Department of Pediatrics, University of South Florida, Box 7523, 880 6th Street South, St. Petersburg, FL 33701. E-mail: estorch@health.usf.edu

Objective: To provide preliminary estimates of the effectiveness of cognitive-behavioral therapy (CBT) in treating anxiety disorders among youth with autism spectrum disorders (ASD) whose anxiety remained clinically problematic following an adequate course of pharmacotherapy.
Method: Seven youth with ASD and one or more comorbid anxiety disorders, who were non- or partial-responders to serotonin reuptake inhibitor (SRI) treatment delivered by their community provider (age range = 12–15 years), received 16 sessions of weekly CBT. Assessments were administered at baseline, approximately 17 weeks following baseline (pre-treatment), and after receiving CBT (post-treatment).
Results: Four of seven participants were classified as treatment responders (much or very much improved) at post-treatment. Clinician severity ratings for 6 of 7 participants—as measured by the PARS, ADIS-IV-P, and CGI-Severity—decreased following CBT, with effect sizes ranging from 1.35 to 1.54. Parent-rated anxiety and ASD symptoms in the child were not significantly reduced.
Conclusions: This study provides preliminary support for use of CBT in augmenting SRI treatment in youth with ASD and anxiety.

Given the high incidence of anxiety in the context of autism spectrum disorders (ASD) (de Bruin, Ferdinand, Meester, de Nijs, & Verheij, 2007; Kaat, Gadow, & Lecavalier, 2013), efforts have been made to develop effective psychosocial interventions. Cognitive-behavioral therapy (CBT) modified for children with ASD and comorbid anxiety has consistently demonstrated efficacy across a number of studies. Although study methodologies vary (e.g., individual versus group, duration of treatment), approximately 60–75% of youth with ASD and anxiety respond to treatment, with corresponding large reductions in anxiety (Reaven, Blakeley-Smith, Culhane-Shelburne, & Hepburn, 2012; Reaven et al., 2009; Sofronoff, Attwood, & Hinton, 2005; Storch et al., 2013; Wood et al., 2009).

Despite the efficacy of CBT for anxiety in youth with ASD, dissemination remains limited and providers often choose pharmacotherapy as a first line of treatment (Coury et al., 2012). Serotonin reuptake inhibitor (SRI) medications are widely used among youth with ASD and comorbid psychopathology such as anxiety, depression, and inattention/hyperactivity (Aman, Lam, & Collier-Crespin, 2003; Langworthy-Lam, Aman, & Van Bourgondien, 2002; Tsiouris, Kim, Brown, Pettinger, & Cohen, 2013). Although controlled efficacy data are limited for SRI medications in youth with ASD and comorbid psychopathology (e.g., anxiety) (Rudy, Lewin, & Storch, 2013; White, Oswald, Ollendick, & Scahill, 2009), some case series-level data are suggestive of benefit (Steingard, Zimnitzky, DeMaso, Bauman, & Bucci, 1997). However, pharmacotherapy interventions may not always be appropriate, desired, or acceptable to parents (Stevens et al., 2009). Further, augmentation approaches, including the potential of sequentially provided CBT to augment non- or partial-response among youth with ASD and anxiety as described in this report, remain untested.

Some evidence supporting the potential usefulness of CBT augmentation may be gleaned from studies in typically developing youth with anxiety. Building on findings among adults with OCD (Simpson et al., 2013; Simpson et al., 2008), Franklin et al. (2011) found that the addition of CBT to a stable antidepressant in youth with OCD was associated with improved response rates (68.6%) relative to youth who received brief instructional CBT together with antidepressant (34.0%) or antidepressant treatment alone (30.0%). Similarly, Storch et al. (Storch, Lehmkuhl et al., 2010) found that the addition of intensive CBT (i.e. daily sessions for three weeks duration) was associated with significantly improved outcomes among youth who remained symptomatic following antidepressant treatment. Two post-hoc studies in non-OCD anxiety suggest that CBT provided sequentially after antidepressant treatment was beneficial for alleviating anxiety symptoms (Eichstedt, Tobon, Phoenix, & Wolfe, 2010; Liashko & Manassis, 2003). Eichstedt et al. (2010) found CBT to be efficacious when delivered after receiving SRI treatment. There were no differences between youth who were taking an SRI (n = 13) versus those who were not on medication (n = 29) following 12 sessions of group CBT at post-treatment and 4-month follow-up with both groups experiencing significant clinical benefit. Similarly, Liashko and Manassis (2003) showed that CBT was effective when augmenting medication treatment; response did not differ between 18 youth taking a psychotropic medication (only 11 of whom were taking an antidepressant) relative to 84 un-medicated children with clinical benefit experienced regardless of medication status. The results of sequential CBT augmentation of SRI treatment of anxiety among typically developing youth are promising and may, therefore, possibly be extended to youth with ASD.

Because SRI medications are widely used in youth with ASD and partial or non-response is common (Nadeau et al., 2011), examination of augmentation strategies are necessary. In the present study, we report case series data examining the merits of developmentally modified CBT for youth with ASD and comorbid anxiety who were sequentially treated with an antidepressant medication prior to receiving 16 CBT sessions. We predicted that participation in CBT would be associated with decreased anxiety symptom severity, functional impairment, depressive, and externalizing symptoms.

METHOD

Participants were 7 youth (5 male) ranging in age from 12 to 15 years who initially presented to a university-based specialty clinic for inclusion in 1 of 3 randomized controlled trials (RCT) evaluating the efficacy of CBT for anxiety in youth with high-functioning ASD (see Table 1 for sample descriptive information). All participants were taking an adequate dose of an SRI for 12 weeks or more with at least 8 weeks at a stable dosage (present dose), deemed by their prescribing physician

TABLE 1
Participant Demographics and Comorbid Diagnoses

Participant	*Gender*	*Ethnicity*	*Age (years)*	*Primary Anxiety Diagnosis*	*ASD Phenotype*	*Comorbid diagnoses*	*Prescribed SRI*	*Dosage (Pre-/ Post-treatment)*
1	Male	White	12	OCD	AS	SoP; GAD	Escitalopram	30mg/30mg
2	Male	Hispanic	12	SoP	AD	GAD; ADHD	Escitalopram	10mg/10mg
3	Male	White	13	SoP	AS	ADHD; SP	Escitalopram	15mg/15mg
4	Male	White	14	SoP	AD	ADHD; SP; GAD	Citalopram	20mg/30mg
5	Male	White	15	SoP	AD	PTSD; ODD; SAD; GAD; Dysthymia; ADHD	Fluoxetine	40mg/50mg
6	Female	White	14	SoP	PDD	SP	Fluoxetine	20mg/20mg
7	Female	White	15	GAD	AD	SoP; ADHD; SP; MDD	Sertraline	25mg/25mg
		Mean (SD)	13.6 (1.3)					

Note: Subjects 2, 3, 4, 5, and 7 received the behavioral interventions for anxiety in children with autism treatment protocol. Subjects 1 and 6 received the anxiety-focused interventions for youth with autism treatment protocol. OCD = Obsessive Compulsive Disorder; SoP = Social Phobia; SP = Specific Phobia; GAD = Generalized Anxiety Disorder; SAD = Separation Anxiety Disorder; MDD = Major Depressive Disorder; AS = Asperger's Syndrome; AD = Autistic Disorder; PDD = Pervasive Developmental Delay, Not Otherwise Specified.

to be of reasonable clinical value and duration based on the youth's age, body mass, and tolerability.

As part of the respective RCTs (Storch et al., 2013), all participants in the current sample were initially randomized to a control condition, either waitlist or treatment as usual (TAU), based on the specific study protocol. Participants randomized to waitlist did not receive additional active treatment (e.g., psychotherapy) for the 16-week waitlist period but remained on their medication without a dosage adjustment. Youth in the TAU condition were able to begin, continue, or change any psychosocial and/or pharmacological treatment for 16 weeks. Two youth increased their SRI dose during the TAU phase; one increased their fluoxetine dose from 40mg to 50mg while the other increased their citalopram from 20mg to 30mg, both approximately 8 weeks after starting the TAU phase. Therefore, collectively, all youth for this study received a stable SRI dosage for 8 weeks or more prior to CBT initiation. After the 16-week control period, youth received 16 weekly CBT sessions per this study's protocol (see the following). Medications were managed by the child's community provider during the 16-week waitlist/TAU period and during their participation in this CBT trial thereafter. Past/current medication use was assessed by parent report at screening, pre-treatment, and post-treatment evaluations ($\sim$17 weeks following baseline). Psychotropic medication dosages remained stable throughout completion of the 16 week CBT protocol (See Table 1 for medication and dosage information).

Additional inclusion criteria for participation included: (a) primary diagnosis of OCD, generalized anxiety disorder, separation anxiety disorder, specific phobia, or social phobia made through the anxiety disorders interview schedule for DSM-IV (ADIS-IV; [Silverman & Albano, 1996]) with a clinical severity rating $\geq$ 4. At the time of study initiation, OCD was still categorized in the *Diagnostic and Statistical Manual for Mental Disorders* (4th edition, text revision [American Psychiatric Association, 2000]) as an anxiety disorder. Although there are phenomenological differences between OCD and other anxiety disorders (McKay & Storch, 2011), OCD was included because: (i) the cognitive behavioral theory of etiology and treatment is similar across disorders, and (ii) the treatment manuals to be utilized address OCD symptoms (Wood et al., in press; Wood & Drahota, 2005) allowing for greater treatment relevance compared to a treatment that assesses a more limited number of problem domains; (b) A diagnosis of ASD made through combined information from the autism diagnosis interview–revised (ADI-R) and autism diagnostic observation schedule–module 3 (ADOS); (c) IQ above 70 on a standardized test. Parents were allowed to produce testing records within the past 2 years documenting that this threshold was achieved; and (d) Availability of at least one parent to attend all treatment sessions with the child. Children were excluded for the following reasons: (i) bipolar disorder, psychotic disorder, or deemed at risk for suicidal behavior; and/or (ii) change in medication within

8 weeks prior to study entry. Study procedures for each RCT were approved by the local Institutional Review Board.

Prior to involvement in each clinical trial, all potential participants were pre-screened via telephone or during regularly scheduled appointments for eligibility; those who appeared to meet study criteria and demonstrated interest were scheduled for a 2-day, in-person screening assessment with each of the days occurring no more than 7 days apart. Written informed assent/consent was obtained from youth and their primary caregiver prior to the screening assessment. Youth who qualified for study participation then returned for a baseline assessment within 1 week of the second screening appointment. Following the waitlist or TAU period, as mentioned above, each of the participants completed a pre-treatment assessment (approximately 17 weeks after the baseline assessment) before initiating CBT as described below. Post-treatment assessments were completed within one week following the final treatment session.

Cognitive-Behavioral Therapy

Participants received 16 weekly sessions of family-based CBT that were 60–90 minutes in length per the behavioral interventions for anxiety in children with autism (BIACA) (Wood & Drahota, 2005) or anxiety-focused interventions for youth with autism (AFIYA) (Wood et al., in press) protocols. The BIACA protocol is an adaptation of traditional CBT for typically developing anxious youth, using a modular treatment approach to address problematic anxiety and non-anxiety symptoms while also considering ASD-specific barriers to care. Each session was divided into two modules (i.e. a parent and a child module), with parents also being present during the child modules as necessary based on individual and developmental needs. BIACA/AFIYA modules were selected prior to each session by the therapist and supervisor in a flexible format to tailor the treatment to each youth's specific needs; however, the core components of traditional CBT were preserved, with at least 3 sessions dedicated to development of coping skills for managing anxiety and 8 sessions dedicated to *in vivo* exposures to feared stimuli. Each of the 5 remaining sessions, as well as portions of the 11 core sessions, may have included skills from 1 or more optional modules such as incorporating positive reinforcement/token economy, communication training, social skills training (9 optional social modules include skills such as friendship, perspective taking, social coaching, etc.), and/or additional repetition of anxiety-focused modules. Additionally, circumscribed interests were utilized for rapport building and reward initially but faded (i.e. targeted for suppression) as treatment progressed. The AFIYA protocol, adapted directly from the BIACA intervention protocol, uses the same modular approach described above with incorporation of more developmentally appropriate language for adolescents and young adults with ASD to serve an older age range. This protocol was used as appropriate with the older youth in the sample.

Doctoral- or post-doctoral-level trainees with concentration in clinical or school psychology and at least 1 year of training and therapeutic experience in CBT for pediatric anxiety disorders served as therapists for the study. Weekly supervision meetings, session content checklists, and selected evaluation of audio-taped sessions were utilized to ensure treatment integrity. Fidelity checks on a 5-point multidimensional scale tailored to fit specific therapy modules revealed excellent: adherence to session components (4.54/5.00), therapist competence (4.75/5.00), therapist flexibility (4.86/5.00), and therapist alliance (4.89/5.00).

Measures

A diagnosis of ASD was established using combined diagnostic information from the ADI-R (Lord, Rutter, & Le Couteur, 1994) and ADOS (Lord, Rutter, DiLavore, & Risi, 1999), obtained during the second screening assessment session by a certified, research-trained, doctoral-level evaluator. A distinction between the diagnosis of autistic disorder and Asperger's disorder was made depending on the presence of a clinically significant language delay in the patient's history consistent with the DSM-IV-TR (American Psychiatric Association, 2000). The ADI-R and ADOS have established psychometrics and demonstrate good diagnostic utility, particularly when used in combination (de Bildt et al., 2004).

Independent evaluators who were not involved in the treatment process administered all clinician-rated measures. Each rater was trained via direct instruction, observation of measure administrations, and completion of measures under supervision prior to study onset. Evaluators were blinded at the baseline and pre-treatment assessments but were not blinded at the post-treatment assessment, given the length of time for follow up.

Pediatric Anxiety Rating Scale (PARS; RUPP, 2002)

The PARS is a clinician-rated scale that assesses anxiety symptoms and the associated severity and impairment in children over the past week based on parent and child endorsement of impairment and distress. The PARS has demonstrated good to excellent psychometrics (e.g., inter-rater reliability, test-retest reliability, convergent validity, divergent validity, and treatment sensitivity) among both typically developing and ASD populations (Storch, Wood et al., 2012). The PARS was used to establish eligibility at screening and was also administered at baseline, pre-treatment, and post-treatment. The intraclass correlation coefficient (ICC) for 19% (4/21 possible assessments) of randomly selected assessments was 0.99.

Anxiety Disorders Interview Schedule for DSM-IV–Child and Parent Versions (ADIS-C/P; Silverman & Albano, 1996)

The ADIS-C/P are semi-structured interviews administered to youth and their parents respectively to assess for the presence of DSM-IV-TR disorders, with a specific emphasis on anxiety disorders. Based on parent and child endorsement of impairment and distress, the ADIS-C/P yields clinician severity ratings (CSR) on 0 to 8 scale with ratings $\geq$ 4 indicating the presence of clinically significant anxiety. The ADIS-C/P demonstrates strong test-retest reliability, inter-rater reliability, and concurrent validity among typically developing youth and has yielded adequate inter-rater agreement (Storch, Ehrenreich May et al., 2012) and treatment sensitivity among youth with ASD (Storch et al., 2013; Wood et al., 2009). The ADIS-C/P was administered at screening, pre-treatment, and post-treatment assessments. Remission was defined as receiving a rating 3 or below (not clinically significant) on the CSR for the primary anxiety diagnosis.

Clinical Global Impression-Severity and -Improvement (CGI-Severity; CGI-Improvement; Guy, 1976)

The CGI-Severity is a single-item rating of illness severity that allows the clinician to rate the global severity of anxiety with scores ranging from 0 ("no illness") to 6 ("serious illness"). The CGI-Improvement is a single-item rating of clinical improvement that allows the clinician to rate improvement in anxiety symptoms from 0 ("very much worse") to 6 ("very much improved"). The CGI-Severity was rated at baseline, pre-treatment, and post-treatment; the CGI-Improvement was rated at pre-treatment and post-treatment. Consistent with others (Storch et al., 2011; Storch, Lewin, De Nadai, & Murphy, 2010), ratings on the CGI-Improvement of "much improved" or "very much improved" were used to define treatment response.

Multidimensional Anxiety Scale for Children – Parent (MASC-P; March, 1998)

The MASC-P is a 39-item measure of general, social, and separation anxiety in children. The MASC-P has demonstrated excellent psychometric properties including internal consistency and test-retest reliability, convergent validity, discriminative validity, and treatment sensitivity with both typically developing and ASD populations (Wood et al., 2009). The MASC-P was administered at baseline, pre-treatment, and post-treatment.

Social Responsiveness Scale (SRS-P; Constantino & Gruber, 2002)

The SRS-P is a 65-item psychometrically sound parent-report measure that assesses symptoms of ASD related to social interaction in youth ages 4 through 18 years. Domains such as social awareness, reciprocal social responses, and autistic preoccupations are rated on a 4-point Likert scale with responses ranging from "never true" (0) to "almost always true" (3). The SRS is able to differentiate between a wide range of social impairment severity and demonstrates excellent psychometric properties (Constantino et al., 2003).

Data Analysis

As examination of data at baseline, pre-treatment, and post-treatment revealed non-normal distributions, non-parametric techniques were used for analysis. Comparison of baseline to pre-treatment scores across all outcome measures revealed no significant differences. Therefore, the pre-treatment time point was utilized for subsequent analyses. The PARS, ADIS-C/P primary anxiety diagnosis CSR, CGI-Severity, CGI-Improvement, MASC-P, and SRS results were analyzed using Wilcoxon signed-rank test (two-sided) comparing pre-treatment and post-treatment time points. Because of the preliminary nature of this study, no correction for family-wise error rate was necessary. To further evaluate the clinical relevance of the findings independent of sample size, the standardized difference between means was calculated as a measure of effect size. The following formula was utilized for this analysis; (Cohen's $d = M_1$ - $M_2/s_{pooled,}$ where s_{pooled} = Ö [(s $_1{}^2$+ s $_2{}^2$)/2] r_{Yl} = d/Ö (d^2 + 4) (Cohen, 1988).

RESULTS

Primary Outcomes

Treatment response was monitored using several outcomes, including the PARS, ADIS-IV CSR, CGI-Severity, CGI-Improvement, and remission status. See Table 2 for a breakdown of treatment response by individual subject, and Table 3 for whole-sample means, standard deviations, inferential statistics, and effect sizes.

Pediatric Anxiety Rating Scale

Comparison of PARS scores showed significant improvement for the overall sample between the pre-treatment and post-treatment assessments, with an average pre-treatment rating of 14.6 ± 3.2 and post-treatment rating of 9.14 ± 4.0, ($T(6) = 3.94$, $p = 0.008$, d = 1.52).

ADIS-IV primary anxiety diagnosis CSR

A comparison of ADIS-IV CSR for each child's primary anxiety diagnosis demonstrated a significant improvement from pre-treatment at the post-treatment assessment, with an average pre-treatment rating of 5.4 ± 1.0 and post-treatment rating of 3.7 ± 1.2, ($T(6) = 3.37$, $p = 0.02$, d = 1.54).

TABLE 2
Individual Participant Treatment Effects

Participant	*Principal Anxiety Diagnosis*	*Baseline CSR*	*Pre-Treatment CSR*	*Post-Treatment CSR*	*Baseline PARS*	*Pre-Treatment PARS*	*Post-Treatment PARS*	*CGI-I*
1	OCD	6	6	5	17	16	14	4
2	Social Phobia	4	4	4	11	11	10	4
3	Social Phobia	6	4	2	13	10	2	6
4	Social Phobia	6	6	0	17	19	11	5
5	Social Phobia	6	6	3	14	16	9	6
6	Social Phobia	6	6	5	15	14	12	4
7	GAD	6	6	3	16	16	6	6

Note: OCD = Obsessive Compulsive Disorder; GAD = Generalized Anxiety Disorder; AS = Asperger's Syndrome; AD = Autistic Disorder; PDD = Pervasive Developmental Delay, Not Otherwise Specified; PARS = Pediatric Anxiety Rating Scale; CSR = Clinician Severity Rating for principal anxiety diagnosis; CGI-I = Clinical Global Impression—Improvement rating.

TABLE 3
Means, Standard Deviations, P-Values, and Effect-Sizes for Outcome Measures

Scale	*Baseline Mean (SD)*	*Pre-Treatment Mean (SD)*	*Effect Size*$^{\Omega}$ *Cohen's d*	*Post-Treatment Mean (SD)*	p-*value (Post-Treatment)*	*Effect Size*$^{\Phi}$ *Cohen's d*
PARS	14.7 (2.2)	14.6 (3.2)	0.05	9.1 (4.0)	0.008	1.52
ADIS CSR	5.7 (0.8)	5.4 (1.0)	0.33	3.7 (1.2)	0.020	1.54
CGI-Severity	3.9 (0.6)	3.7 (1.1)	0.23	2.4 (0.8)	0.022	1.35
MASC-P	61.0 (11.2)	60.6 (14.0)	0.03	55.3 (10.5)	Ns (0.091)	0.43
SRS	83.9 (6.8)	80.0 (7.3)	0.55	77.3 (11.3)	Ns (0.394)	0.28

Note: SD = Standard deviation; PARS = Pediatric Anxiety Rating Scale; ADIS CSR = Anxiety Disorders Interview Schedule for DSM-IV Principal Anxiety Clinician Severity Rating; CGI-Severity = Clinical Global Impression—Severity rating; MASC-P = Multidimensional Anxiety Scale for Children—Parent Version; SRS = Social Responsiveness Scale; Ns = not significant.

$^{\Omega}$Effect sizes were calculated based on Baseline- and Pre-Treatment differences.

$^{\Phi}$Effect sizes were calculated based on Pre- and Post-Treatment differences.

CGI-severity

The sample, as a whole, demonstrated a positive treatment response on the CGI-Severity, with an average pre-treatment rating of 3.7 ± 1.1 and post-treatment rating of 2.4 ± 0.8, ($T(6) = 3.06$, $p = 0.02$, d = 1.35). Correspondingly, 6 of 7 children demonstrated at least 1 grade of improvement on the CGI-Severity at the post-treatment assessment point. At post-treatment, anxiety severity on the CGI-Severity was classified as "slight illness" for 1 (Subject 3), "mild" for 2 (Subjects 5 and 7), and "moderate" for 4 (Subjects 1, 2, 4, and 6).

CGI-improvement

At post-treatment, 4 of 7 children were considered treatment responders, defined as rater endorsement of "much improved" or "very much improved".

Remission status

At post-treatment, the primary anxiety diagnosis of 4 participants (Subjects 3, 4, 5, and 7) were classified as in remission, defined as scoring 3 or below (not clinically significant) on the ADIS-IV-CSR.

Secondary Outcomes

Multidimensional Anxiety Scale for Children-Parent and Social Responsiveness Scale

Scores on the MASC-P and SRS were not significantly different from pre-treatment at the post-treatment assessment (MASC-P: $T(6)$ 2.01, $p = 0.09$, d = 0.43; SRS: $T(6)$ 0.92, $p = 0.39$, d = 0.28).

DISCUSSION

This case series provides preliminary support that CBT modified for youth with ASD may be beneficial for those who present with clinically significant anxiety after antidepressant treatment. Due in part to limited CBT dissemination for anxious youth with and without ASD, antidepressants are frequently used as an initial treatment among youth with comorbid anxiety and ASD (Aman et al., 2003; Langworthy-Lam et al., 2002; Tsiouris et al., 2013). However, partial and non-response to pharmacotherapy is common (King et al., 2009), and strategies for augmentation (e.g., sequential CBT) are therefore necessary. Overall, the data from this case series are encouraging, as the cohort showed an average PARS reduction of 37% corresponding with a large effect size. More than half of the sample qualified as treatment responders, and 86% (6 of 7 participants) demonstrated some level of improvement.

Although most children gleaned some benefit, only 4 of 7 (57%) participants achieved clinical remission for their primary anxiety diagnosis, and scores remained in the elevated range for overall anxiety ratings. This is not surprising, as youth with ASD may demonstrate somewhat lower treatment response than their typically developing anxious peers (Storch et al., 2013). Overall, the degree of symptom reduction was similar to that found in previously conducted RCTs examining modified CBT for youth with comorbid anxiety and ASD (Reaven et al., 2012; Reaven et al., 2009; Sofronoff et al., 2005; Storch et al., 2013; Wood et al., 2009). Children with ASD often demonstrate diagnostic complexity, increased likelihood of comorbidity, and a host of psychosocial and family factors that complicate the therapeutic process and symptom alleviation (Nadeau et al., 2011). That said, the results of this case series suggest that modified CBT is a viable option for sequential augmentation of pharmacotherapy to address comorbid anxiety among the pediatric ASD population; however, more research is necessary to determine the most promising treatment methods in terms of treatment response.

Several limitations should be noted. First, the absence of a comparison control condition makes it difficult to determine the specific effects of CBT relative to other interventions. Second, the medication regimens were managed by community physicians rather than as a formal and controlled aspect of the study protocol. As such, pharmacological regimens may or may not have always been based on published practice guidelines, and there was no way to ascertain if the medication dosage was optimized. Third, a range of antidepressants was used, and dosages were assessed through parent report without external verification. Fourth, the small sample size utilized in this preliminary report precluded examining predictors, moderators, or mediators of response. For example, we were not able to assess the extent to which ADHD comorbidity was associated with attenuated outcomes, which has been demonstrated previously in typically developing youth (Storch et al., 2008) and was clinically noted in this sample to potentially impact treatment delivery. Fifth, outside of ASD specific measures (e.g., SRS), the study measurement battery included anxiety instruments that were not specifically designed for youth with ASD. On balance, several of the measures have demonstrated psychometric properties in this population (Storch, Ehrenreich May et al., 2012; Storch, Wood et al., 2012). However, we highlight the importance of future studies to employ measures of anxiety designed specifically for children with ASD (Sukhodolsky et al., 2008), as well as employ a multi-method assessment given issues in symptom reporting in youth with ASD (Storch, Ehrenreich May et al., 2012; White, Schry, & Maddox, 2012). Finally, no follow-up assessment was conducted to determine the durability of treatment gains. Despite these limitations, several modest design strengths should be noted. First, medication dosages were stable prior to screening, and control conditions across the relevant studies maintained or increased dosages prior to the 16-week active CBT phase. Thus, youth had an adequate duration of pharmacotherapy prior to the provision of CBT. Second, our measurement model was rigorous, affording a comprehensive assessment of anxiety and related symptoms. Third, the active treatment (CBT) utilized empirically supported CBT treatment protocols for treating anxiety in youth with ASD that were tailored to individual participants yet administered in a standardized fashion, ensuring an adequate and appropriate dose of psychotherapy.

Practice Implications

These preliminary data suggest that CBT modified for youth with ASD and comorbid anxiety may be beneficial in addressing anxiety symptoms among children who remained symptomatic following pharmacotherapy. Currently, practice guidelines for youth with ASD and anxiety have not been standardized as have those in typically developing youth with anxiety (Connolly & Bernstein, 2007; Geller & March, 2012). Studies of individual and group CBT have consistently demonstrated efficacy in reducing anxiety among youth with ASD (Reaven et al., 2012; Storch et al., 2013; Wood et al., 2009; Wood et al., in press), yet dissemination remains limited at this time. Pharmacotherapy with antidepressants (e.g., SRIs) is commonly used (Tsiouris et al., 2013), although the evidence to support this approach for anxiety in youth with ASD is limited. Although firm conclusions cannot be made given the open trial nature and small sample size, the current results provide initial support for the utility of CBT to augment pharmacotherapy among this population.

FUNDING

The project was supported in part by National Institute of Child Health and Development award R34HD065274 to Dr. Storch. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institute of Child Health and Development or the National Institutes of Health.

REFERENCES

Aman, M. G., Lam, K. S., & Collier-Crespin, A. (2003). Prevalence and patterns of use of psychoactive medicines among individuals with autism in the Autism Society of Ohio. *Journal of Autism and Developmental Disorders*, *33*, 527–534.

American Psychiatric Association. (2000). *Diagnostic and statistical manual of mental disorders* (4th ed., text rev.). Washington, DC: Author.

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences*. Hillsdale, NJ: Erlbaum.

Connolly, S. D., & Bernstein, G. A. (2007). Practice parameter for the assessment and treatment of children and adolescents with anxiety disorders. *Journal of the American Academy of Child and Adolescent Psychiatry*, *46*, 267–283.

Constantino, J. N., Davis, S. A., Todd, R. D., Schindler, M. K., Gross, M. M., Brophy, S. L., . . . Reich, W. (2003). Validation of a brief quantitative measure of autistic traits: Comparison of the social responsiveness scale with the autism diagnostic interview-revised. *Journal of Autism and Developmental Disorders*, *33*, 427–433.

Constantino, J. N., & Gruber, C. P. (2002). *The social responsiveness scale*. Los Angeles, CA: Western Psychological Services.

Coury, D. L., Anagnostou, E., Manning-Courtney, P., Reynolds, A., Cole, L., McCoy, R., . . . Perrin, J. M. (2012). Use of psychotropic medication in children and adolescents with autism spectrum disorders. *Pediatrics, 130 Supplement 2*, S69–76.

de Bildt, A., Sytema, S., Ketelaars, C., Kraijer, D., Mulder, E., Volkmar, F., & Minderaa, R. (2004). Interrelationship between Autism Diagnostic Observation Schedule-Generic (ADOS-G), Autism Diagnostic Interview-Revised (ADI-R), and the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV-TR) classification in children and adolescents with mental retardation. *Journal of Autism and Developmental Disorders*, *34*, 129–137.

de Bruin, E. I., Ferdinand, R. F., Meester, S., de Nijs, P. F., & Verheij, F. (2007). High rates of psychiatric co-morbidity in PDD-NOS. *Journal of Autism and Developmental Disorders*, *37*, 877–886.

Eichstedt, J. A., Tobon, J. I., Phoenix, E., & Wolfe, V. V. (2010). Worried no more: The effects of medication status on treatment response to a CBT group for children with anxiety in a community setting. *Clinical Child Psychology and Psychiatry*, *16*, 265–277.

Franklin, M. E., Sapyta, J., Freeman, J. B., Khanna, M., Compton, S., Almirall, D., . . . March, J. S. (2011). Cognitive behavior therapy augmentation of pharmacotherapy in pediatric obsessive-compulsive disorder: The pediatric OCD treatment study II (POTS II) randomized controlled trial. *JAMA*, *306*, 1224–1232.

Geller, D. A., & March, J. (2012). Practice parameter for the assessment and treatment of children and adolescents with obsessive-compulsive disorder. *Journal of the American Academy of Child and Adolescent Psychiatry*, *51*, 98–113.

Guy, W. (1976). Clinical global impressions. *ECDEU Assessment Manual for Psychopharmacology*. (vol. revised DHEW Pub., pp. 218–222). Rockville, MD: National Institute for Mental Health.

Kaat, A. J., Gadow, K. D., & Lecavalier, L. (2013). Psychiatric symptom impairment in children with autism spectrum disorders. *Journal of Abnormal Child Psychology*, *41*, 959–969.

King, B. H., Hollander, E., Sikich, L., McCracken, J. T., Scahill, L., Bregman, J.D., . . . Ritz, L. (2009). Lack of efficacy of citalopram in children with autism spectrum disorders and high levels of repetitive behavior: Citalopram ineffective in children with autism. *Archives of General Psychiatry*, *66*, 583–590.

Langworthy-Lam, K. S., Aman, M. G., & Van Bourgondien, M. E. (2002). Prevalence and patterns of use of psychoactive medicines in individuals with autism in the Autism Society of North Carolina. *Journal of Child and Adolescent Psychopharmacology*, *12*, 311–321.

Liashko, V., & Manassis, K. (2003). Medicated anxious children: characteristics and cognitive-behavioural treatment response. *Canadian Journal of Psychiatry*, *48*, 741–748.

Lord, C., Rutter, M., DiLavore, P. C., & Risi, S. (1999). *Autism diagnostic observation schedule*. Los Angeles, CA: Western Psychological Services.

Lord, C., Rutter, M., & Le Couteur, A. (1994). Autism diagnostic interview-revised: A revised version of a diagnostic interview for caregivers of individuals with possible pervasive developmental disorders. *Journal of Autism and Developmental Disorders*, *24*, 659–685.

March, J. (1998). *Manual for the Multidimensional Anxiety Scale for Children*. Toronto, Canada: Multi-Health Systems Inc.

McKay, D., & Storch, E. A. (2011). *Handbook of anxiety disorders in children and adolescents*. New York, NY: Springer Publishing Company.

Nadeau, J., Sulkowski, M. L., Ung, D., Wood, J. J., Lewin, A., Murphy, T., . . . Storch, E. A. (2011). Treatment of anxiety in patients with autism spectrum disorders. *Neuropsychiatry*, *1*, 567–578.

Reaven, J., Blakeley-Smith, A., Culhane-Shelburne, K., & Hepburn, S. (2012). Group cognitive behavior therapy for children with high-functioning autism spectrum disorders and anxiety: A randomized trial. *Journal of Child Psychology and Psychiatry*, *53*, 410–419.

Reaven, J., Blakeley-Smith, A., Nicols, S., Dasari, M., Flanigan, E., & Hepburn, S. (2009). Cognitive-behavioral group treatment for anxiety symptoms in children with high-functioning autism spectrum disorders, a pilot study. *Focus on Autism and Other Developmental Disabilities*, *24*, 27–37.

Rudy, B. M., Lewin, A., & Storch, E. A. (2013). Managing anxiety comorbidity in youth with autism spectrum disorders. *Neuropsychiatry*, *3*, 411–421.

RUPP. (2002). The Pediatric Anxiety Rating Scale: Development and psychometric properties. *Journal of the American Academy of Child and Adolescent Psychiatry*, *41*, 1061–1069.

Silverman, W. K., & Albano, A. M. (1996). *The anxiety disorders interview schedule for DSM-IV–Child and parent versions*. San Antonio, TX: Graywinds Publications.

Simpson, H. B., Foa, E. B., Liebowitz, M. R., Huppert, J. D., Cahill, S., Maher, M. J., . . . Campeas, R. (2013). Cognitive-behavioral therapy vs risperidone for augmenting serotonin reuptake inhibitors in obsessive-compulsive disorder: A randomized clinical trial. *JAMA Psychiatry, 70*, 1190–1199.

Simpson, H. B., Foa, E. B., Liebowitz, M. R., Ledley, D. R., Huppert, J. D., Cahill, S., . . . Petkova, E. (2008). A randomized, controlled trial of cognitive-behavioral therapy for augmenting pharmacotherapy in obsessive-compulsive disorder. *American Journal of Psychiatry*, *165*, 621–630.

Sofronoff, K., Attwood, T., & Hinton, S. (2005). A randomised controlled trial of a CBT intervention for anxiety in children with Asperger syndrome. *Journal of Child Psychology and Psychiatry*, *46*, 1152–1160.

Steingard, R. J., Zimnitzky, B., DeMaso, D. R., Bauman, M. L., & Bucci, J. P. (1997). Sertraline treatment of transition-associated anxiety and agitation in children with autistic disorder. *Journal of Child and Adolescent Psychopharmacology*, *7*, 9–15.

Stevens, J., Wang, W., Fan, L., Edwards, M. C., Campo, J. V., & Gardner, W. (2009). Parental attitudes toward children's use of antidepressants and psychotherapy. *Journal of Child and Adolescent Psychopharmacology*, *19*, 289–296.

Storch, E. A., Arnold, E. B., Lewin, A., Nadeau, J., Jones, A. M., de Nadai, A. S., . . . Murphy, T. (2013). The effect of cognitive-behavioral therapy versus treatment as usual for anxiety in children with Autism Spectrum Disorders: A randomized, controlled trial. *Journal of the American Academy of Child and Adolescent Psychiatry, 52*, 132–142.

Storch, E. A., Ehrenreich May, J., Wood, J.J., Jones, A.M., de Nadai, A.S., Lewin, A., . . . Murphy, T.K. (2012). Multiple informant agreement on the Anxiety Disorders Interview Schedule in youth with autism spectrum disorders. *Journal of Child and Adolescent Psychopharmacology*, *22*, 292–299.

Storch, E. A., Lehmkuhl, H. D., Ricketts, E., Geffken, G. R., Marien, W., & Murphy, T. K. (2010). An open trial of intensive family based cognitive-behavioral therapy in youth with obsessive-compulsive disorder who are medication partial responders or nonresponders. *Journal of Clinical Child and Adolescent Psychology*, *39*, 260–268.

Storch, E. A., Lewin, A. B., De Nadai, A. S., & Murphy, T. K. (2010). Defining treatment response and remission in obsessive-compulsive disorder: A signal detection analysis of the Children's Yale-Brown Obsessive Compulsive Scale. *Journal of the American Academy of Child and Adolescent Psychiatry*, *49*, 708–717.

Storch, E. A., Merlo, L. J., Larson, M. J., Geffken, G. R., Lehmkuhl, H. D., Jacob, M. L., . . . Goodman, W. K. (2008). Impact of comorbidity on cognitive-behavioral therapy response in pediatric obsessive-compulsive disorder. *Journal of the American Academy of Child and Adolescent Psychiatry*, *47*, 583–592.

Storch, E. A., Wood, J. J., Ehrenreich-May, J., Jones, A. M., Park, J. M., Lewin, A., & Murphy, T.K. (2012). Convergent and discriminant validity and reliability of the Pediatric Anxiety Rating Scale in youth with autism spectrum disorders. *Journal of Autism and Developmental Disorders, 42*, 2374–2382.

Sukhodolsky, D. G., Scahill, L., Gadow, K. D., Arnold, L. E., Aman, M. G., McDougle, C. J., . . . Vitiello, B. (2008). Parent-rated anxiety symptoms in children with pervasive developmental disorders: Frequency and association with core autism symptoms and cognitive functioning. *Journal of Abnormal Child Psychology*, *36*, 117–128.

Tsiouris, J. A., Kim, S. Y., Brown, W. T., Pettinger, J., & Cohen, I. L. (2013). Prevalence of psychotropic drug use in adults with intellectual disability: Positive and negative findings from a large scale study. *Journal of Autism and Developmental Disorders*, *43*, 719–731.

White, S. W., Oswald, D., Ollendick, T., & Scahill, L. (2009). Anxiety in children and adolescents with autism spectrum disorders. *Clinical Psychology Review*, *29*, 216–229.

White, S. W., Schry, A. R., & Maddox, B. B. (2012). The assessment of anxiety in high-functioning adolescents with autism spectrum disorder. *Journal of Autism and Developmental Disorders*, *42*, 1138–1145.

Wood, J. J., & Drahota, A. (2005). *Behavioral interventions for anxiety in children with autism*. Los Angeles, CA: UCLA.

Wood, J. J., Drahota, A., Sze, K., Har, K., Chiu, A., & Langer, D. A. (2009). Cognitive behavioral therapy for anxiety in children with autism spectrum disorders: A randomized, controlled trial. *Journal of Child Psychology and Psychiatry*, *50*, 224–234.

Wood, J. J., Ehrenreich May, J., Alessandri, M., Fujii, C., Renno, P., Laugeson, E., . . . Storch, E. A. (2015). Cognitive behavioral therapy for early adolescents with autism spectrum disorders and clinical anxiety: A randomized, controlled trial. *Behavior Therapy, 46*(1), 7–19.